INTE 22253 Distributed Systems and Cloud Computing

Introduction to Cloud Computing

From Basics to Modern Developments

Lecture 2

Lecture Agenda

01 The World of Data & Big Data

02 What is Cloud Computing?

03 Cloud Service Models (IaaS, PaaS, SaaS)

04 Cloud Deployment Models

05 Enabling Technologies

06 Cloud Characteristics & Benefits

07 Modern Cloud Developments (2020–2025)

08 Edge Computing & Fog Computing

09 Serverless & Containers

10 AI/ML in the Cloud & Future Trends

Our Data Now…



Documents & Files

Word, Excel, PDFs, Presentations



Media

Photos, Videos, Music, Podcasts



Communications

Emails, Chats, Video Calls



Personal Data

Calendar, Contacts, Location



Health Data

Wearables, Medical Records



Financial Data

Banking, Transactions, Investments

The World of Data

2.5

Quintillion Bytes

created every day

5B+

Internet Users

worldwide (2024)

500M+

Tweets

sent daily

720K hrs

YouTube Video

uploaded every day

We live in an unprecedented era of data generation — and this drives the need for cloud computing.

Big Data

Big data refers to large pools of data that can be captured, communicated, aggregated, stored, and analyzed to deliver insights and value.

Volume	Velocity	Variety	Veracity	Value
From 1.2 ZB (2010) to 120+ ZB (2024)	Real-time streaming & processing	Structured, semi-structured & unstructured	Data quality & accuracy	Business insights & decisions

The 5 V's of Big Data

What Do We Do With Data?



Store

Databases, file systems, object storage



Secure

Encryption, access control, compliance



Share

APIs, collaboration platforms



Process

Real-time analytics, batch processing



Analyze

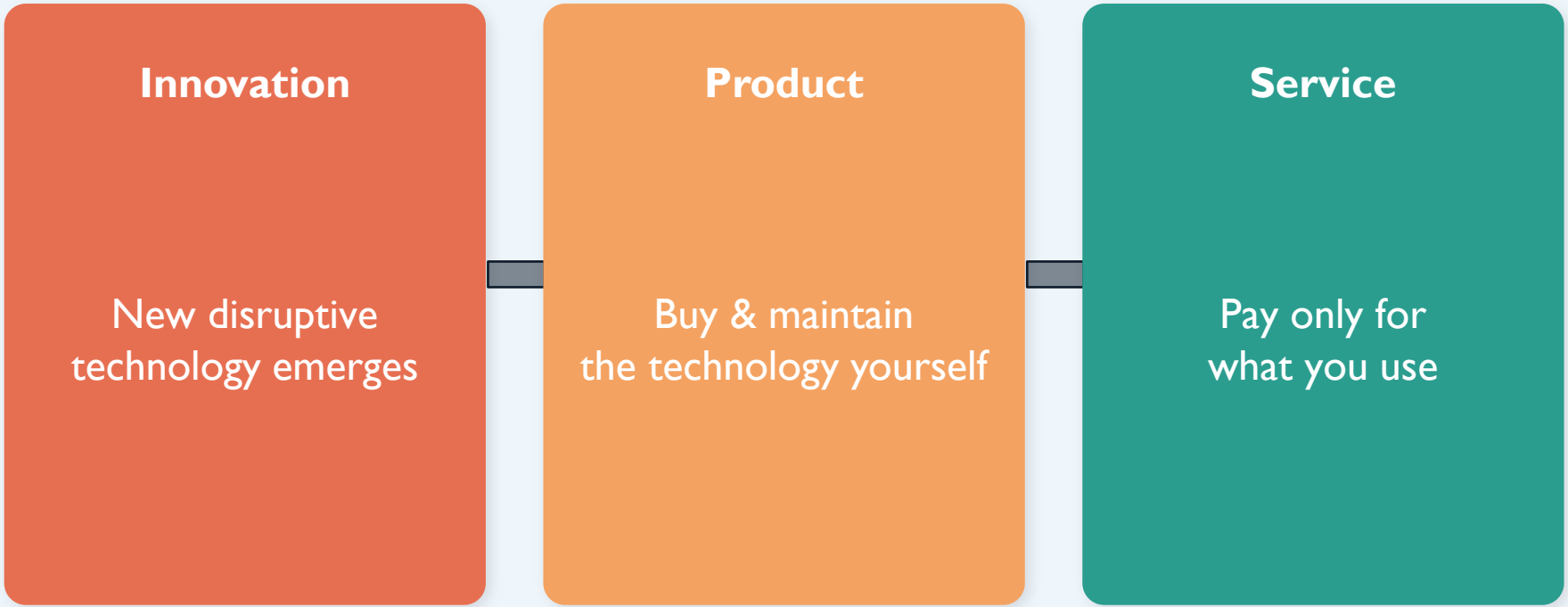
ML, AI, BI dashboards



Stream

IoT sensors, live data pipelines

Has This Happened Before?



Electricity, Water, Banking — all transformed from products to services. Computing follows the same path!

The Electricity Analogy



Then

Companies generated their own electricity with private generators. Expensive, unreliable, required specialists.



Transition

Electricity became a shared utility — buy from the grid, pay only what you use, via a reliable connection.



Now: Cloud

Companies access computing on-demand. No upfront servers. Pay per use. Managed by experts.

So… What is Cloud Computing?

"Cloud Computing is the delivery of computing as a service rather than a product, whereby shared resources, software, and information are provided to computers and other devices as a metered service over a network."

= On-Demand IT Services on a Pay-as-You-Go basis over the Internet

Think of it like electricity: plug in and pay only for what you use.

Cloud Computing — Key Concepts

Remotely Hosted	Services and data are stored and run on remote infrastructure, not on local machines.
Ubiquitous Access	Data and services accessible from anywhere, anytime, on any device.
Commodified Computing	Computing as a utility — like water or electricity. Pay only for what you consume.
Multi-Tenancy	Multiple users share the same physical infrastructure securely and independently.

NIST's 5 Essential Characteristics

1

On-Demand Self-Service

Users provision compute resources automatically without human interaction from the provider.

2

Broad Network Access

Available over the network through standard mechanisms (phones, tablets, laptops).

3

Resource Pooling

Provider's resources are pooled to serve multiple consumers using multi-tenant model.

4

Rapid Elasticity

Capabilities can be elastically provisioned and released to scale rapidly with demand.

5

Measured Service

Resource usage can be monitored, controlled, and reported — pay for what you use.

Why Cloud Computing?

💰 Pay-as-You-Go

No upfront capital expenditure. Reduce Total Cost of Ownership.

📈 Scale Instantly

Resources can be rented and released as needed. Instant scalability.

🔧 Simplified IT

Provider manages the infrastructure. All you need is internet access.

🌎 Flexible Options

Configure software, OS, platforms. Access from anywhere with internet.

♻️ Resource Efficiency

Reduce idle resources by sharing. Better CPU/storage utilization.

🌿 Green Computing

Sharing resources = fewer servers, less power, lower carbon footprint.

Cloud Service Models



IaaS

Infrastructure as a Service

Provides virtualized computing infrastructure (VMs, storage, networking). You manage OS and above.

Examples: AWS EC2, Azure VMs, Google Compute Engine



PaaS

Platform as a Service

Provides a platform for developers to build and deploy applications. You manage only the app.

Examples: Heroku, Google App Engine, AWS Elastic Beanstalk



SaaS

Software as a Service

Complete software delivered over the internet. No installation needed. Pay-as-you-go.

Examples: Gmail, Salesforce, Microsoft 365, Zoom

IaaS — Infrastructure as a Service

What You Manage

- Applications
- Data
- Runtime
- Middleware
- OS

Provider Manages

- Virtualization
- Servers
- Storage
- Networking

Best for: Lift-and-shift migrations, development/test environments, HPC workloads

PaaS — Platform as a Service

Developers focus only on writing code. The platform handles everything else.

⚙️ Auto-scaling

Automatically handle traffic spikes

🔄 CI/CD Pipelines

Automated build, test, deploy

🗄️ Managed Databases

No DBA needed — provider manages backups

📦 Dev Frameworks

Pre-built runtime environments

🌐 Built-in Networking

Load balancers, SSL certificates handled

📊 Monitoring

Built-in logging and performance tools

Examples: Heroku, Google App Engine, Azure App Service, AWS Elastic Beanstalk, Firebase

SaaS — Software as a Service

Ready-to-use applications delivered over the internet — no installation required.

Email

Gmail, Outlook 365

Video Conf.

Zoom, Teams, Meet

CRM

Salesforce, HubSpot

Office Suite

Microsoft 365, Google Workspace

Media

Spotify, Netflix

Security

Okta, CrowdStrike

Analytics

Tableau Online, Power BI

Collaboration

Slack, Notion, Asana

Beyond IaaS/PaaS/SaaS — The 'as a Service' Revolution

FaaS

Function as a Service

Run individual functions on-demand. AWS Lambda, Azure Functions.

DBaaS

Database as a Service

Managed databases — no DBA. AWS RDS, Azure SQL, MongoDB Atlas.

STaaS

Storage as a Service

Scalable object/file storage. S3, Azure Blob, Google Cloud Storage.

AIaaS

AI as a Service

Pre-trained ML models via API. AWS SageMaker, Azure AI, Google Vertex.

NaaS

Network as a Service

Virtual networking on-demand. Software-Defined Networking (SDN).

SECaaS

Security as a Service

Cloud-based security — SIEM, CASB, DLP. CrowdStrike, Zscaler.

Cloud Deployment Models

🌐 Public Cloud

Resources owned and operated by a third-party provider, delivered over the internet.

Pros: Low cost, high scalability, no maintenance
Examples: AWS, Azure, Google Cloud

🏢 Private Cloud

Cloud infrastructure operated exclusively for a single organization.

Pros: High security, compliance, control
Examples: VMware, OpenStack

🔀 Hybrid Cloud

Combination of public and private clouds, bound together by technology enabling data and application portability.

Pros: Flexibility, cost-optimized

☁️☁️ Multi-Cloud

Using multiple public cloud providers simultaneously.

Pros: Avoid vendor lock-in, best-of-breed services
2024: Used by 87% of enterprises

Enabling Technologies for Cloud Computing

🌐 Internet & Web 2.0

High-bandwidth, always-on connectivity makes remote computing feasible.

⚡ Virtualization

Allows physical hardware to be divided into multiple virtual machines, maximizing utilization.

⚙️ Parallel & Distributed Computing

Splitting workloads across many machines enables massive scale.

💡 Utility Computing

Pay-per-use billing model mirrors traditional utilities — you pay for what you consume.

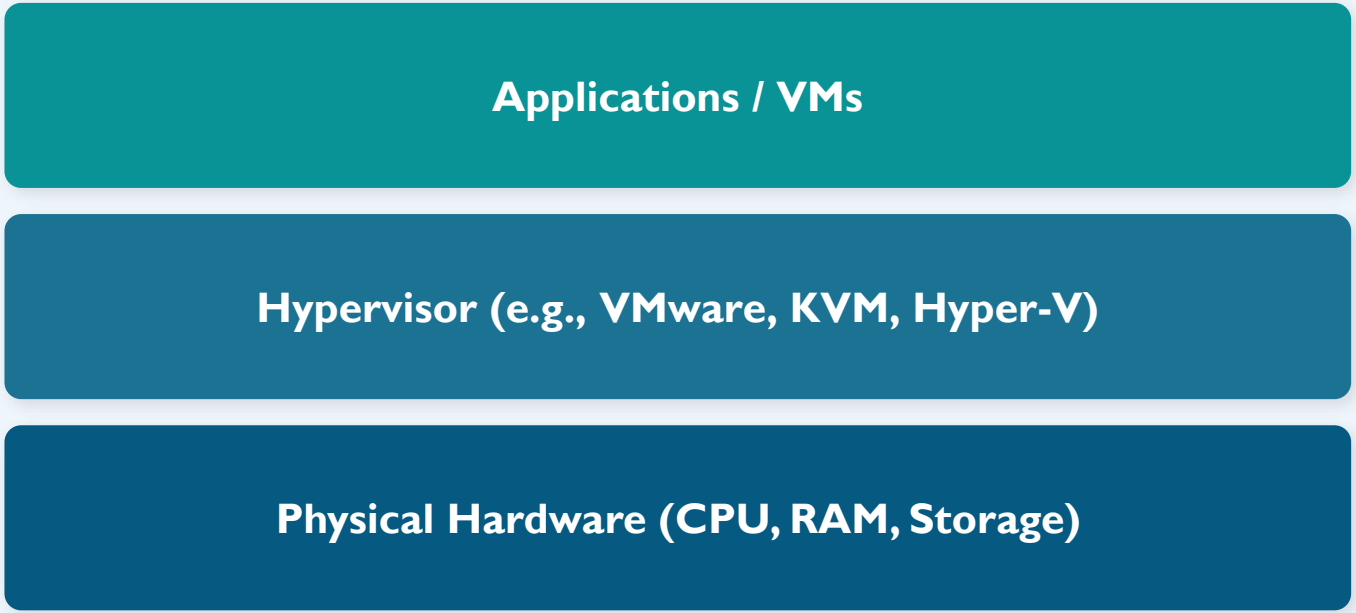
💾 Storage Technologies

Object storage, distributed file systems (HDFS, GCS) enable petabyte-scale data.

🛡️ Fault Tolerance

Redundancy and replication ensure services stay up even when components fail.

Virtualization — The Core Enabler



Key Benefits of Virtualization:

- Server consolidation
- Isolation between workloads
- Rapid VM provisioning
- Snapshotting & migration

Cloud Computing Characteristics

▸ Massive Scale
Globally distributed data centers running millions of servers.

▸ Homogeneity
Standardized hardware enables consistent management at scale.

▸ Resilient Computing
Auto-healing systems recover from failures automatically.

▸ Geographic Distribution
Data centers across regions for low latency and compliance.

▸ Low-Cost Software
Open-source stacks reduce licensing costs dramatically.

▸ Advanced Security
Shared responsibility model — provider secures infrastructure.

▸ Service Orientation
Everything exposed as an API — composable, interoperable.

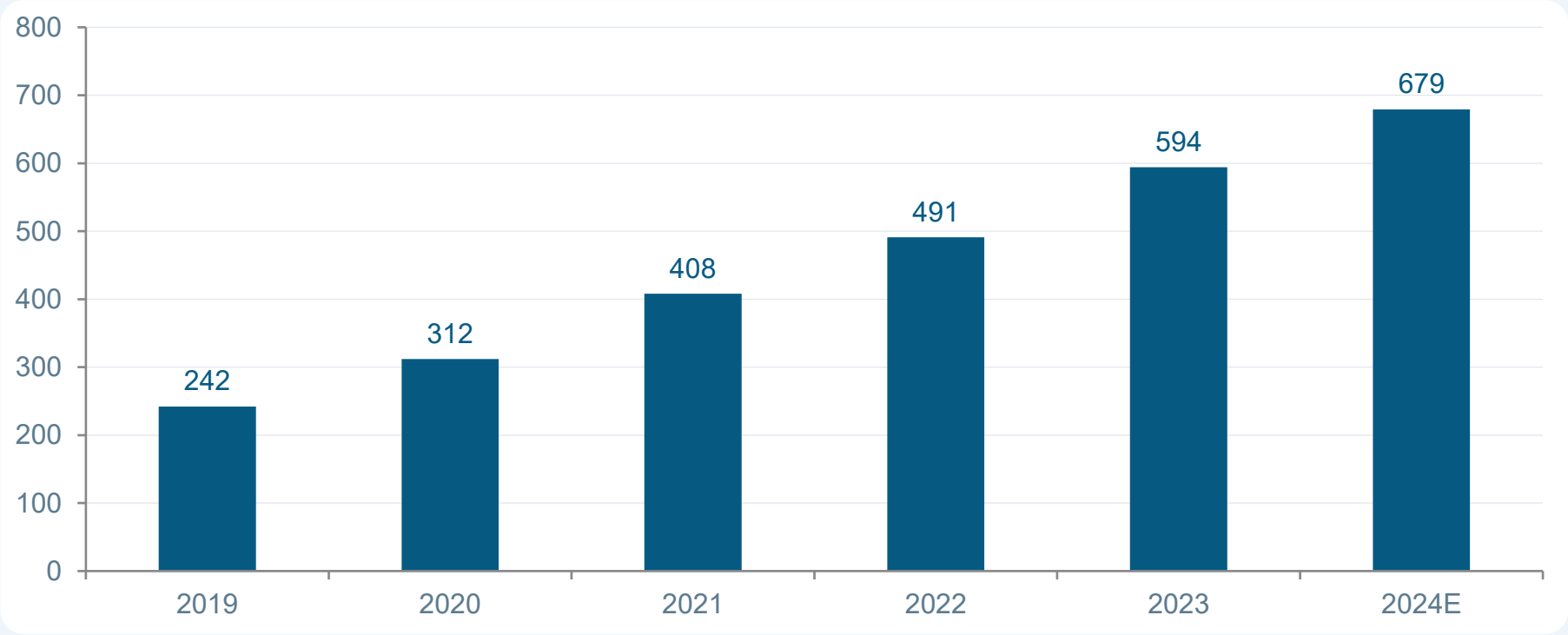
▸ Virtualization
Abstracts physical resources into flexible logical resources.

Major Cloud Providers (2024)

AWS	**Amazon Web Services** EC2, S3, Lambda, RDS, SageMaker, 200+ services	Market Share: 31%
Azure	**Microsoft Azure** VMs, Blob Storage, Azure Functions, Cosmos DB	Market Share: 25%
GCP	**Google Cloud** Compute Engine, BigQuery, GKE, Vertex AI	Market Share: 11%
Alibaba	**Alibaba Cloud** Dominant in Asia-Pacific region	Market Share: 4%

Source: Synergy Research Group, Q3 2024

Cloud Market — Explosive Growth



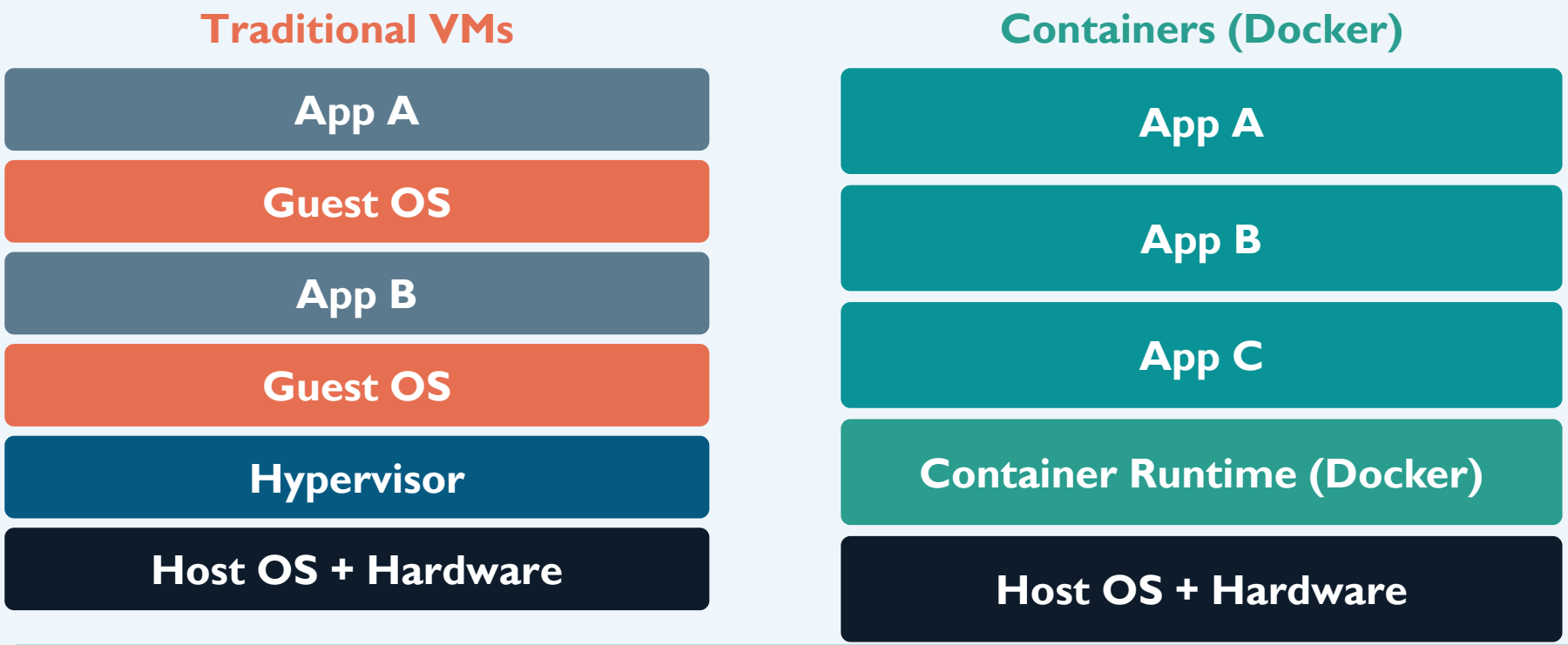
Global public cloud services revenue has grown 3× in 5 years. Expected to exceed $1 trillion by 2027.

Modern Cloud Developments

2020 – 2025 and Beyond

- 🔲 **Containers & Kubernetes**
- ⚡ **Serverless Computing**
- 🌿 **Edge & Fog Computing**
- 🤖 **AI/ML as a Service**
- 🔒 **Zero Trust Security**
- 🌍 **Sustainability & Green Cloud**

Containers & Docker



Containers share the host OS — they are faster, lighter, and more portable than VMs.

Kubernetes — Container Orchestration

Kubernetes (K8s) automates deployment, scaling, and management of containerized applications.

⚙ Auto-Scaling

Scale pods up/down based on CPU, memory, or custom metrics automatically.

⚙ Self-Healing

Restarts failed containers and kills containers that don't respond to health checks.

⚙ Load Balancing

Distributes network traffic to pods to ensure stable deployment.

⚙ Rolling Updates

Deploy new versions with zero downtime; roll back instantly if needed.

⚙ Service Discovery

Automatically assigns DNS names and IP addresses to containers.

⚙ Storage Orchestration

Automatically mounts storage systems — local, cloud, or network.

Managed K8s: Amazon EKS · Azure AKS · Google GKE

Serverless Computing (FaaS)

"Serverless" doesn't mean no servers — it means developers don't manage servers. The cloud provider provisions, scales, and manages infrastructure automatically.

Event Trigger (HTTP, Timer, Queue) → **Function Executes (milliseconds)** → **Auto-Scales (0 to millions)** → **Pay for Execution Only**

- ✅ **No server management**
- ✅ **Auto-scales to zero**
- ✅ **Pay only per invocation**
- ✅ **Fast deployment**
- ⚠️ Cold start latency
- ⚠️ Debugging is harder

AWS Lambda · Azure Functions · Google Cloud Functions · Cloudflare Workers

Edge Computing

Processing data closer to where it is generated — at the 'edge' of the network — rather than in a centralized data center.



- 🚗 Autonomous Vehicles — Real-time decisions cannot wait for cloud round-trip latency.
- 🏭 Industrial IoT — Manufacturing sensors process locally for immediate control.
- 🏥 Remote Healthcare — Medical devices process vitals at the edge for instant alerts.
- 🎮 Cloud Gaming — Edge servers deliver ultra-low latency gaming experiences.

Fog Computing & MEC

Cloud Computing

- Centralized data centers
- High latency (50–150ms)
- Massive computational power
- Ideal for batch analytics, storage
- Examples: AWS, Azure, GCP

Fog Computing

- Between edge & cloud
- Medium latency (10–50ms)
- Aggregates edge data
- Cisco's term for distributed compute
- Examples: OpenFog, Cisco IOx

Edge Computing

- On-device or gateway
- Ultra-low latency (<10ms)
- Lightweight, real-time compute
- Processes at the source
- Examples: AWS Wavelength, Azure Edge

AI & Machine Learning in the Cloud

Cloud is the engine of the AI revolution — training large models requires massive distributed compute.

AWS · SageMaker

End-to-end ML platform: build, train, deploy

Azure · Azure AI / OpenAI

GPT-4, DALL-E, Cognitive Services

Google · Vertex AI

AutoML, model garden, Gemini integration

All · GPU Instances

NVIDIA A100/H100 GPUs for deep learning training

Generative AI spending in the cloud is projected to exceed $200B annually by 2028.

Generative AI & Cloud Computing

The relationship between GenAI and cloud is symbiotic:

🤖 **Foundation Models**	GPT-4, Claude, Gemini, LLaMA require thousands of GPUs — only feasible in cloud.
☁️ **AI Infrastructure**	Cloud providers built specialized AI chips (TPUs, Trainium, Inferentia) for cost efficiency.
🔌 **AI APIs**	Developers access AI capabilities via simple REST APIs — no ML expertise needed.
🏢 **Enterprise AI**	Companies fine-tune foundation models on their private data using cloud platforms.
🔒 **Data Privacy**	Cloud providers offer VPC, private endpoints for AI model training on sensitive data.
📊 **Vector Databases**	New cloud-native vector DBs (Pinecone, Weaviate, pgvector) power AI applications.

Cloud Security

The Shared Responsibility Model

Security OF the Cloud (Provider)

Physical infrastructure · Network hardware · Hypervisor

Security IN the Cloud (Customer)

Data encryption · Access management (IAM) · Network configuration · Application security

🔒 **Zero Trust Security: 'Never trust, always verify' — the modern cloud security**

Zero Trust Security Architecture

Traditional perimeter security is obsolete in the cloud. Zero Trust assumes breach and verifies continuously.

🔍 **Verify Explicitly**	Authenticate and authorize based on all available data points — identity, location, device health, workload.
🔑 **Least Privilege**	Limit user and workload access to only what's needed, at the right time, for the right purpose.
💥 **Assume Breach**	Design as if attackers are already in the network. Minimize blast radius and segment access.

Tools: Microsoft Entra ID · Zscaler · CrowdStrike · Palo Alto Networks SASE

Cloud Cost Optimization (FinOps)

Cloud costs can spiral without management. FinOps is the practice of bringing financial accountability to cloud spending.

💡 Right-Sizing

Match instance size to actual workload. Oversized instances waste money.

💡 Reserved Instances

Commit to 1-3 year reservations for predictable workloads — save up to 72%.

💡 Spot/Preemptible VMs

Use spare cloud capacity for fault-tolerant workloads at 60-90% discount.

💡 Auto-Scaling

Scale down during low demand. Don't pay for idle resources overnight/weekends.

💡 Storage Tiering

Move infrequently accessed data to cheaper storage tiers (Glacier, Archive).

💡 Cost Allocation Tags

Tag every resource — track spending by team, project, or environment.

Compliance, Governance & Data Sovereignty

GDPR

EU General Data Protection Regulation. Strict rules on personal data — right to erasure, data residency.

HIPAA

US Health Insurance Portability Act. Healthcare data must be encrypted and access-logged.

PCI-DSS

Payment Card Industry standard. Controls for systems that process credit card data.

ISO 27001

International information security management standard — widely adopted for cloud.

SOC 2

Service Organization Controls. Trusted cloud providers achieve Type I/II attestations.

PDPA / DPDPA

Personal Data Protection Acts — emerging regulations in Sri Lanka, India & Southeast Asia.

Sustainability & Green Cloud Computing

Data centers consume ~1-2% of global electricity. Cloud providers are racing to achieve net-zero emissions.

Google Cloud	Carbon neutral since 2007. Targeting 24/7 carbon-free energy by 2030.
Microsoft Azure	Carbon negative by 2030. Water positive by 2030. 100% renewable energy.
AWS	Committed to 100% renewable energy by 2025. Net-zero carbon by 2040.

🌿 **Cloud consolidation is inherently greener: shared infrastructure has 65-80% better energy efficiency than individual on-premise servers**

Multi-Cloud & Hybrid Cloud Strategy

87% of enterprises use multiple cloud providers. Why?

1. Avoid Vendor Lock-in — Not dependent on a single provider's pricing, features, or outages.
2. Best-of-Breed — Use AWS for compute, GCP for AI/ML, Azure for Microsoft integrations.
3. Regulatory Compliance — Some regions require data to stay in-country — use local cloud provider.
4. Cost Arbitrage — Price competition between providers; use cheapest for each workload.
5. Redundancy — If one cloud has an outage, workloads can failover to another provider.

Tools: Terraform, Pulumi, Kubernetes (cloud-agnostic), HashiCorp Vault

DevOps & Cloud-Native Development

Cloud-native apps are designed specifically to exploit cloud capabilities from the ground up.

Microservices

Decompose apps into small independent services. Each scales independently.

CI/CD Pipelines

Automated build, test, deploy — release multiple times per day safely.

Infrastructure as Code

Manage infra via code (Terraform, CloudFormation). Version-controlled, repeatable.

Observability

Logs, metrics, traces — understand system behavior in production (Datadog, Grafana).

Chaos Engineering

Deliberately inject failures to test resilience (Netflix Chaos Monkey).

GitOps

Git as single source of truth for infrastructure. Auto-sync cluster state to git.

Real-World Cloud Case Studies

Netflix AWS	Migrated from own data centers to AWS. Streams to 260M+ subscribers globally. Uses 100K+ AWS instances at peak.
Airbnb AWS	Scaled from startup to $75B valuation on cloud. Handles 150M+ user listings with no data center costs.
Spotify Google Cloud	Migrated to GCP for BigQuery analytics. Processes 600GB of data daily for music recommendations.
BMW AWS + Azure	Multi-cloud for connected cars. 3M+ connected vehicles send real-time telemetry data to the cloud.

Cloud Computing in Sri Lanka & South Asia

🏛️ Government Digitization

Sri Lanka's ICT Agency (ICTA) pursuing cloud-first policy. Lanka Government Cloud (LGC) initiative.

🏦 Banking Sector

Major banks (Commercial Bank, HNB) using hybrid cloud for core banking modernization.

📱 Telecom & Connectivity

Dialog, SLT offering cloud services to SMEs. Improving broadband penetration for cloud access.

🎓 Education

Universities moving to Microsoft 365, Google Workspace. E-learning platforms on cloud.

🌏 Regional Data Centers

AWS, Azure have Singapore, India regions. Sub-10ms latency available for Sri Lankan enterprises.

📊 Startups & Fintech

Growing startup ecosystem leveraging AWS/GCP to scale globally without local infrastructure.

Future Trends in Cloud Computing

Quantum Computing in Cloud

2025–2030

IBM, Google, AWS offering quantum computers via cloud. Will break current encryption — need quantum-resistant crypto.

AI-Ops & Autonomous Cloud

2024–2026

AI managing cloud infrastructure — auto-healing, predictive scaling, anomaly detection with no human intervention.

Confidential Computing

2024–2025

Encrypted computation — data is processed while encrypted. Enables secure multi-party computation.

6G & Cloud Integration

2028–2030

6G networks will enable seamless edge-to-cloud continuum with terabit speeds and sub-millisecond latency.

Neuromorphic Computing

2027–2032

Brain-inspired chips for ultra-efficient AI inference. Intel Loihi chips offered as cloud services.

Space-Based Cloud

2025–2030

Starlink, AWS Kuiper, Azure Orbital bringing cloud connectivity to remote areas via satellite.

Cloud Certifications — Career Pathways

AWS

- Cloud Practitioner (Beginner)
- Solutions Architect Associate
- Developer Associate
- DevOps Engineer Professional

Microsoft Azure

- AZ-900: Azure Fundamentals
- AZ-104: Administrator
- AZ-204: Developer
- AZ-305: Solutions Architect

Google Cloud

- Cloud Digital Leader
- Associate Cloud Engineer
- Professional Cloud Architect
- Professional ML Engineer

Certifications significantly boost employability — average certified cloud engineer salary: $120K–$180K (US)

Knowledge Check — Test Yourself!

1. What are the 5 essential characteristics of cloud computing according to NIST?

→ *On-demand self-service, broad network access, resource pooling, rapid elasticity, measured service.*

2. Explain the shared responsibility model in cloud security.

→ *Provider secures infrastructure; customer secures data, apps, access, and network config within the cloud.*

3. What is the difference between a container and a virtual machine?

→ *VMs virtualize hardware with a full guest OS; containers share the host OS kernel — lighter and faster.*

4. When would you choose serverless over containers?

→ *Serverless: event-driven, infrequent, short-lived functions. Containers: long-running, stateful, complex apps.*

Lecture Summary

1	**Data explosion drives demand:**	We generate 2.5 quintillion bytes daily — cloud provides infrastructure to store, process & analyze this.
2	**Cloud = IT as a utility:**	Just like electricity, computing transformed from a product to an on-demand, metered service.
3	**3 service models:**	IaaS (infrastructure), PaaS (platform), SaaS (software) — plus FaaS, DBaaS, AIaaS and more.
4	**4 deployment models:**	Public, Private, Hybrid, Multi-Cloud — choose based on cost, control, compliance needs.
5	**Modern developments:**	Containers, Kubernetes, Serverless, Edge Computing, AI/ML cloud, Zero Trust security.
6	**Sustainability matters:**	Cloud consolidation reduces carbon footprint; major providers targeting net-zero by 2030.

Lecture 2 Complete

Questions?

📚 **Next Lecture: Virtualization in Depth**

Topics: Hypervisors (Type 1 & 2), Hardware virtualization, Memory virtualization, SR-IOV, Live migration

📖 **Recommended Reading:**

Cloud Computing: Concepts, Technology & Architecture — Thomas Erl | AWS / Azure / GCP Free Tier — Try it today!

Uditha Dharmakeerthi